

John the Ripper — Practical Tool Guide

Wordlists, formats, --show, sessions, zip2john, SSH key workflow, and how to identify a format instead of guessing.

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Usage note: Offline cracking for lab/authorized data only. Do not use it on credential material you are not authorized to process.

EXTRACT

hash / file

IDENTIFY

format

CRACK

wordlist

SHOW

result

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1. John concept

John works offline on hashes or hashes extracted from files. Identify what you have before starting.

KALI TERMINAL — Help / formats

```
john --help
john --list=formats | less
```

KALI TERMINAL — Wordlist attack

```
john --wordlist=/usr/share/wordlists/rockyou.txt [HASH_FILE]
```

KALI TERMINAL — Known format

```
john --format=[FORMAT] --wordlist=[WORDLIST] [HASH_FILE]
```

KALI TERMINAL — Show results

```
john --show [HASH_FILE]
john --show --format=[FORMAT] [HASH_FILE]
```

2. How do I know the format?

Use the source/context and identification helpers, then verify against John formats.

KALI TERMINAL — Identification clue

```
hashid [HASH_FILE]
```

Note: hashid is a clue, not a final decision.

KALI TERMINAL — Search John formats

```
john --list=formats | grep -i "nt\|bcrypt\|sha512"
```

Often the source tells you more than the string: NTLM commonly comes from Windows; SHA512crypt often begins with \$6\$. Do not invent --format values.

3. Files that need *2john

ZIP, SSH, PDF, and other protected files usually need to be converted to a hash format John understands first.

KALI TERMINAL — ZIP

```
zip2john protected.zip > zip.hash  
john --wordlist=[WORDLIST] zip.hash  
john --show zip.hash
```

KALI TERMINAL — SSH private key

```
ssh2john id_rsa > ssh.hash  
john --wordlist=[WORDLIST] ssh.hash  
john --show ssh.hash
```

In some Jumbo/Kali packages the utility may be named ssh2john.py or live under a package path. Check the installed package.

KALI TERMINAL — Find converters

```
ls /usr/share/john/*2john* 2>/dev/null | head -40
```

4. Sessions and pot file

John saves cracked results so work does not need to be repeated unnecessarily.

KALI TERMINAL — Named session

```
john --session=lab1 --wordlist=[WORDLIST] [HASH_FILE]
```

KALI TERMINAL — Restore

```
john --restore=lab1
```

KALI TERMINAL — Show instead of re-cracking

```
john --show [HASH_FILE]
```

5. Workflow

Step	Action	Goal
1	Source	Know where the hash/file came from.
2	Convert	Use *2john for ZIP/SSH/PDF when required.
3	Identify	Use context + hashid + formats.
4	Attack	Start with a wordlist.
5	Show	Display the result.
6	Validate	Test the credential only inside lab scope.

Prepared as a quick-reference companion to the canonical Arabic guide. Use only in labs or systems you are explicitly authorized to test.

References:

<https://www.openwall.com/john/doc/>

<https://github.com/openwall/john>

Openwall John Jumbo documentation: *2john utilities